HANDSHAKE

PROJECT NAME:	Receiver and Transmitter using NRF24L Module and STM Microcontroller
PROJECT GROUP:	Sahil Khandare & Pratik Khadse

WHY THIS PROJECT ?

- This project was undertaken to create an alternative to the existing Flysky transmitter– receiver system. Our goal is to design and develop a custom transmitter and receiver tailored specifically to our requirements. These requirements will evolve based on the future competitions the club participates in.
- Our mission is to develop and build all electronic components in-house, ensuring full control over performance, customization and innovation.

INTRODUCTION

- This project is developed using an STM microcontroller paired with the NRF24L01 wireless module. Both the PCB and the firmware were designed and built entirely in-house. The NRF24L01 was selected primarily for its high data rate, which supports fast and efficient communication between the transmitter and receiver. However, during development, we realized that this choice came with a trade-off—our focus on speed led us to overlook the module's limited operational range.

KEY FEATURES

1. Custom Firmware Development:
   Entire firmware written in-house, allowing complete control over optimization.
2. In-House PCB Development:
   Custom-designed PCB tailored specifically for the transmitter–receiver architecture.
3. High Data Rate Communication:
   Utilizes the NRF24L01 module's fast data-rate capability for efficient and low-latency communication.

ARCHITECTURE

1. Microcontroller Integration:
   Rx - STM32G070RBT6
   Tx - STM32G030KBT6
2. Wireless Module Selection:
   NRF24L01
3. PCB Material :
   Copper Clad using Acid Etching.

RESULTS

- The custom transmitter–receiver system successfully established stable communication using the STM microcontroller and NRF24L01 module, with the high data rate delivering fast and responsive performance.
- The in-house PCB fabricated through acid-wash etching on copper-clad boards operated reliably, supporting clean signal routing and consistent power delivery. However, testing revealed a notable limitation in communication range, as the emphasis on high data rate reduced overall distance performance.

FUTURE SCOPE

1. ☐ **Use of HC-12**
2. ☐ **Multi Rx with Single Tx**
3. ☑ **Going for OSHWA certification**

PAPERWEIGHT

PROJECT NAME:	Custom Flight Controller {An Attempt with Learning}
PROJECT GROUP:	Sahil Khandare & Pratik Khadse

WHY THIS PROJECT ?

- This project was undertaken to create an alternative Flight Controller. Our goal is to design and develop a Custom Flight Controller specifically to our requirements. These requirements will evolve based on the future competitions the club participates in.
- Our mission is to develop and build all electronic components in-house, ensuring full control over performance, customization and innovation.

INTRODUCTION

- This project was developed using an STM microcontroller integrated with sensors such as an IMU and a barometer. The flight controller PCB was designed and fabricated entirely in-house, and ChibiOS was selected for firmware development. However, during firmware flashing, the board failed due to the absence of proper ESD protection, which led to damage to the microcontroller and prevented successful programming.

KEY FEATURES {WE THOUGHT}

1. ChibiOS-Based Firmware Architecture
   The firmware was planned to be developed using ChibiOS on the SpeedyBee F405 .
2. In-House PCB Development:
   Custom-designed PCB tailored specifically for the future expansion.
3. Sensor Integration:
   The controller was designed to interface with an IMU and barometric sensor to provide accurate attitude estimation, altitude measurement, and stable flight control.

ARCHITECTURE

1. Microcontroller Integration:
   STM32F405RGT6
2. Barometer Selection:
   BMP200
3. IMU Selection :
   ICM 42670

RESULTS

- The project provided valuable hands-on learning in custom flight controller design, including hardware integration, PCB development, and real-time firmware planning. Although firmware flashing failed due to the absence of ESD protection, this highlighted the critical importance of robust hardware protection, proper grounding, and design-for-reliability practices. Overall, the experience strengthened understanding of embedded system design challenges and informed improvements for future flight controller revisions.

FUTURE SCOPE

1. **Another Attempt**

ROBOTICS CONTROL UNIT

PROJECT NAME:	ROBOTICS CONTROL UNIT
PROJECT GROUP:	SAHIL KHANDARE, PRATIK KHADSE & VEDANTI DAMDHAR

WHY THIS PROJECT ?

- This project was undertaken to develop a custom ROBOTICS CONTROL UNIT for driving motors according to specific system requirements. The objective is to design and build an in-house motor driver that can be tailored and optimized as performance needs evolve in future competitions undertaken by the club.

INTRODUCTION

- The project was to be used to replace the L298 series Motor Driver being used at our club.
- The whole Design and Development was done for Line follower and RC Car.

KEY FEATURES

1. Dual Motor output capable of 2 amps of current.
2. Cortex M0+ MCU @ 64MHz with 32kB RAM and 128kB FLASH which can be used for data logging as well.
3. Multiple input for sensors, 1 UART and 1 I2C Channel.

ARCHITECTURE

1. Microcontroller Integration:
   STM32G070KBT6
2. DRIVER Selection:
   DRV8837 BY TEXAS INSTUMENTS

RESULTS

- The development of the motor driver for the Robotics Control Unit demonstrated the feasibility of in-house motor control electronics for robotic applications. The project resulted in a functional design approach for driving motors with controlled speed and direction, while emphasizing the importance of protection circuitry, power handling, and signal integrity. Through this process, practical insights were gained into motor control principles, hardware reliability, and system-level integration, which will directly inform improved and more robust motor driver designs in future iterations.

FUTURE SCOPE

1. ☐ **Closed Loop Control**
2. ☐ **ESD Protection**
3. ☐ **PID Tuning and Field Oriented Control for smoother control**

3D-PRINTED QUAD-DRONE

PROJECT NAME:	3D-Printed Quad-Drone
PROJECT GROUP:	Sahil Khandare, Pratiksha Bawaskar & Ishali Umale

WHY THIS PROJECT ?

- First Person View (FPV) drones are widely used in racing, inspection, research and defense-related applications. However, learning FPV flight on high-cost commercial drones often leads to damage and financial loss during training.
- This project was undertaken to develop a low-cost and mass producible specifically intended for training and flight-testing purposes. By using a 3D-printed frame, the design can be rapidly modified, repaired, or reprinted, making it ideal for beginners and experimental testing.

INTRODUCTION

- The 3D-Printed FPV Drone is a lightweight quadcopter designed primarily to test FPV flight characteristics and to train new pilots.
- The frame is fully 3D printed, allowing design flexibility and quick iteration. The drone serves as a testbed platform for experimenting with different FPV setups, flight controller tuning, and payload arrangements.

KEY FEATURES

- 3D-Printed Airframe enables rapid prototyping, customization, and low-cost replacement.
- FPV Training Platform
- Flight Testing Platform
- Used to test FPV flight dynamics before integrating FPV into advanced drones.
- Allows easy replacement and upgrading of components.
- Cost-Effective Development
- Significantly reduces risk and expense during pilot training.

ARCHITECTURE

- Use of 3D Printing
- 3D printing was chosen to:
- Reduce manufacturing cost
- Allow fast design iterations
- Enable lightweight structural designs
- Simplify repairs by reprinting damaged parts
- The frame design can be modified based on flight performance and pilot feedback.

RESULTS

- The 3D-printed FPV drone successfully demonstrated stable flight and effective FPV operation
- Minor vibrations were observed due to the lightweight printed structure, which provided valuable data for future design improvements.

FUTURE SCOPE

1. **Frame reinforcement using composite or carbon-infused filaments**
2. **Improved vibration damping techniques**
3. **Use as a swarm or multi-drone training platform**

DRONE CHARGING STATION

PROJECT NAME:	Drone Charging Station (Metal Contact-Based System)
PROJECT GROUP:	Pavan Kajale, Riddhi Ambere

WHY THIS PROJECT ?

- This project focuses on the development of a contact-based drone charging station that enables automatic and safe battery charging using a metal contact system. In this approach, the UAV lands on a charging pad equipped with metal contact plates.
- The system allows the drone to recharge autonomously without manual cable connection, thereby improving operational efficiency and enabling long-duration or repetitive missions.

INTRODUCTION

- The 3D-Printed FPV Drone is a lightweight quadcopter designed primarily to test FPV flight characteristics and to train new pilots.
- The frame is fully 3D printed, allowing design flexibility and quick iteration. The drone serves as a testbed platform for experimenting with different FPV setups, flight controller tuning, and payload arrangements.

KEY FEATURES

- 3D-Printed Airframe enables rapid prototyping, customization, and low-cost replacement.
- FPV Training Platform
- Flight Testing Platform
- Used to test FPV flight dynamics before integrating FPV into advanced drones.
- Allows easy replacement and upgrading of components.
- Cost-Effective Development
- Significantly reduces risk and expense during pilot training.

- Use of 3D Printing
- 3D printing was chosen to:
- Reduce manufacturing cost
- Allow fast design iterations
- Enable lightweight structural designs
- Simplify repairs by reprinting damaged parts
- The frame design can be modified based on flight performance and pilot feedback.

RESULTS

- The 3D-printed FPV drone successfully demonstrated stable flight and effective FPV operation
- Minor vibrations were observed due to the lightweight printed structure, which provided valuable data for future design improvements.

FUTURE SCOPE

1. **Frame reinforcement using composite or carbon-infused filaments**
2. **Improved vibration damping techniques**
3. **Use as a swarm or multi-drone training platform**

SYSTEM FOR MAPPING OF FLOOD AND EARLY WARNING CONFIGURATION BY MEANS OF AN UAV

PROJECT NAME:	FLOOD PREDICTION SYSTEM
PROJECT GROUP:	Hrishikesh Kakade & Saurabh Gaikwad

WHY THIS PROJECT ?

- Floods cause major loss of life and property and traditional monitoring methods are often slow and limited in coverage. This project aims to provide rapid and accurate flood assessment using UAV technology.
- Early warning through aerial data helps authorities take timely action and improve disaster preparedness.

INTRODUCTION

- This project involves the use of an Unmanned Aerial Vehicle (UAV) and Satellite images to map flood-affected areas and monitor water levels in real time.
- The system collects aerial images and aerial videos to support flood analysis and early warning generation

KEY FEATURES

- Creates a 3D simulation model to give the visual view of the water level.
- Early warning system based on water level data and flood extent analysis.
- Divide the zones for construction areas and provides to Government authorities for Early Planning.

ARCHITECTURE

- The UAV is equipped with cameras, sensors, GPS, and a communication module for data acquisition.
- Collected data is transmitted to a ground station where it is processed for mapping and alert generation.

RESULTS

- The UAV successfully captured high-resolution images of River side areas for accurate mapping of flood.
- The system demonstrated effective early warning capability by detecting rising water levels

FUTURE SCOPE

1. **Integration of AI and machine learning for automatic flood prediction and damage assessment**
2. **Expansion to large-scale deployment with multiple UAVs for continuous monitoring of river basins and urban areas.**

VIRTUAL FENCING & NAVIGATION SYSTEM FOR AUTONOMOUS ROVER

PROJECT NAME:	Virtual fencing & navigation system for autonomous mobile rover
PROJECT GROUP:	Pranav Harne & Atharva Ingle

WHY THIS PROJECT ?

- Physical fencing is costly, inflexible, and difficult to modify for changing environments. This project provides a software-based virtual fencing solution that allows easy boundary updates without physical infrastructure
- It reduces human supervision by enabling autonomous navigation within safe, predefined area.

INTRODUCTION

- This project presents an autonomous mobile rover that navigates within virtual boundaries using positioning and sensor data.
- The system ensures safe movement, obstacle avoidance, and controlled operation without the need for physical barriers.

KEY FEATURES

- Virtual fencing (geo-fencing) restricts the rover's movement within defined zones.
- Autonomous navigation with real-time obstacle detection ensures safe and efficient operation.

ARCHITECTURE

- The rover consists of sensors (GPS, ultrasonic/IR), a microcontroller, motor drivers, and a power system.
- Navigation algorithms process boundary data and sensor inputs to control rover movement and decision-making.

RESULTS

- The rover successfully operated within the virtual fence without crossing restricted areas.
- Accurate navigation and stable movement were achieved in real-time test environments

FUTURE SCOPE

1. **Warehouse automation**
2. **Smart surveillance systems**
3. **It can be applied in agriculture, security patrols**

SMART GUARDIAN

PROJECT NAME:	Smart Guardian
PROJECT GROUP:	Vedika Raut

WHY THIS PROJECT ?

- The Smart Guardian project is developed to enhance safety and security by automatically detecting threats like human intrusion and fire, reducing the need for continuous human monitoring.
- The main objective is to enhance security and protection by continuously monitoring critical parameters and responding instantly to abnormal conditions.

INTRODUCTION

- Smart Guardian is an IoT-based intelligent surveillance and safety system that monitors surroundings in real time using camera and sensors and provides instant alerts during emergencies.
- The system continuously senses, analyzes, and transmits data to ensure timely alerts and preventive action.

KEY FEATURES

- Real-time video streaming
- Fire detection using smoke and temperature sensors
- Instant alert notifications
- YOLO-based human detection
- Allows easy replacement and upgrading of components.
- Low-cost and scalable system

ARCHITECTURE

- The system uses an ESP32-CAM to capture and stream live video.
- The video stream is processed using Python-based YOLO for real-time human detection.
- Fire detection is achieved using MQ smoke sensor and DHT11 temperature sensor to monitor smoke concentration and temperature rise.
- Alerts are sent to the user through internet-based notification services.

RESULTS

- The Smart Guardian system successfully detects human presence and fire conditions in real time and sends timely alerts, ensuring improved safety and surveillance.

FUTURE SCOPE

1. **Cloud-based data storage , AI-based threat prediction**
2. **Deployment in smart homes, industries, and drones**
3. **Integration of face recognition**

S.O.S SYSTEM

PROJECT NAME:	Rescue Alert System as Human Safety Device
PROJECT GROUP:	Nivrutti Raut & Parth Deshmukh

WHY THIS PROJECT ?

- The Rescue Alert System is developed to enhance human safety during emergencies by providing a fast and reliable alert mechanism
- The main goal is to design a customized safety device on clothes that can instantly transmit distress signals to rescue teams or guardians..

INTRODUCTION

- The Rescue Alert System is a human safety device designed using an Arduino and GSMA7670C with GPS NEO 6M.
- The system enables quick transmission of emergency alerts during critical situations.

KEY FEATURES

1. Custom Firmware Development
   - Embedded firmware is developed in-house for real-time alert generation and transmission.
   -
2. In-House PCB Design
   - Customized PCB designed specifically for safety-device requirements, ensuring compactness and reliability.
   -
3. Emergency Alert Mechanism
   - Allows instant triggering of alerts during danger situations.

ARCHITECTURE

1. Microcontroller: ATmega328P
2. Wireless Communication Module: GSMA7670C
3. User Interface / Trigger Mechanism: Push Button
4. Location Access: GPS NEO 6M

RESULTS

- The Rescue Alert System successfully demonstrated quick and reliable emergency communication.
- The system showed response , stable wireless transmission, and consistent performance.
- The system proves effective as a human safety device for emergency rescue applications.

FUTURE SCOPE

1. ☐ **Integration of GPS for Location Tracking**
2. ☐ **Multi-Receiver Communication**
3. ☐ **Certification and Deployment for Public Safety Use**

SEED SPREADER

PROJECT NAME:	Multi-Purpose Agriculture Drone – Seed Spreader
PROJECT GROUP:	CLASS OF 2024

WHY THIS PROJECT ?

- Traditional farming methods are time-consuming and labor-intensive.
- Manual seed spreading leads to uneven distribution and wastage.
- Farmers face difficulties accessing large or uneven farmland.
- Supports smart farming and precision agriculture

INTRODUCTION

- The Multi-Purpose Agriculture Drone is an unmanned aerial vehicle (UAV) developed for agricultural applications.
- It is primarily used for seed spreading, and can also support fertilizer spraying and field inspection.

KEY FEATURES

- Seed spreading mechanism
- Controlled seed distribution
- GPS-assisted stable flight (optional)
- Easy operation with remote controller
- Covers large areas in less time

ARCHITECTURE

- Carbon fiber Hexacopter frame design for stability
- BLDC motors : HOBBYWING X8 Motor
- Flight Controller : Jiyi K++ V2
- Li-Po Battery : Pair of Tattu 22000mAh
- Tank : 16 Litres Capacity

RESULTS

- The agriculture drone successfully demonstrated controlled flight and seed spreading.
- Significant reduction in manual effort and time consumption.
- Stable flight operation with smooth, controlled, and reliable UAV movement by continuously correcting disturbances

FUTURE SCOPE

1. **Integration of AI-based crop analysis**
2. **Autonomous navigation using GPS**
3. **Multi-payload system for spraying and monitoring**

MULTI-PURPOSE AGRICULTURE DRONE

PROJECT NAME:	Multi-Purpose Agriculture Drone
PROJECT GROUP:	CLASS OF 2024

WHY THIS PROJECT ?

- Manual spraying is time-consuming, inefficient, and labor-intensive.
- Uneven distribution leads to wastage of water, seeds, or fire retardants.
- Hard-to-reach or hazardous areas are difficult to cover safely.
- Drones provide precise, fast, and safe spraying for agriculture, reforestation, and firefighting.

INTRODUCTION

- This project focuses on using drones for spraying water, seeds, or fire retardants efficiently over large or hard-to-reach areas.
- It reduces manual labor, improves precision, and ensures safer and faster spraying operations.

KEY FEATURES

- Efficient spraying of water, seeds, fertilizer, or flower distribution
- Covers plants, flowers, large fields, and hard-to-reach areas
- Reduces manual labor and effort
- Ensures precise and uniform distribution
- Supports safe and faster operations

ARCHITECTURE

- Carbon fiber Hexacopter frame design for stability
- BLDC motors : HOBBYWING X8 Motor
- Flight Controller : Jiyi K++ V2
- Li-Po Battery : Pair of Tattu 22000mAh

RESULTS

- Drone successfully performed spraying operations for water, seeds, fire retardants, and flowers.
- Materials were distributed evenly and precisely over the target area.
- Reduced manual labor and increased efficiency in spraying tasks

FUTURE SCOPE

1. ☐ • **Integration of AI-based crop analysis**
2. ☐ **Autonomous navigation using GPS**
3. ☐ **Multi-payload system for spraying and monitoring**

FPV DRONE RACING

PROJECT NAME:	FPV Drone Racing – 3D Printed Propeller Guard
PROJECT GROUP:	CLASS OF 2024

WHY THIS PROJECT ?

- FPV racing drones operate at high speed, increasing crash risk.
- Propellers are the most damage-prone and dangerous components.
- Crashes can cause injury, propeller breakage, and motor damage.
- Commercial propeller guards are costly and not customizable.
- 3D printing allows lightweight, low-cost, and replaceable safety guards.

INTRODUCTION

- FPV drone racing requires high agility and rapid maneuvers, often leading to collisions.
- The 3D Printed Propeller Guard is designed to protect propellers, motors, and surroundings without affecting flight performance significantly.
- The guard is custom-designed and fabricated using 3D printing, allowing easy modification.
- It improves safety during training, beginner practice, and close-proximity racing.

KEY FEATURES

- Lightweight 3D printed design
- Protects propellers from direct impact
- Reduces injury risk to pilots and spectators
- Custom-fit for FPV racing frames
- Easy installation and replacement
- Improves durability of racing drone

ARCHITECTURE

- The flight controller acts as the main control unit of the drone.
- The power system supplies energy to all electronic components.
- Motors and propellers generate lift and movement.
- The radio system allows the pilot to control the drone wirelessly.
- The FPV system provides live video transmission to the pilot.

RESULTS

- The FPV drone successfully flew with the 3D printed propeller guard installed.
- The guard provided effective protection during minor crashes.
- Minimal impact on flight stability and maneuverability.
- Improved confidence during training sessions.

FUTURE SCOPE

1. **Use of flexible TPU filament for better shock absorption**
2. **Aerodynamic optimization to reduce drag**
3. **Integration with racing frames**

AEROVISION

PROJECT NAME: Aerovision – UAV-Based Aerial Survey and Vision based data acquisition

PROJECT GROUP: Atharva Raut

WHY THIS PROJECT ?

- Traditional aerial survey systems are expensive, complex, and manpower-dependent
- Real-time monitoring in remote, disaster-prone, and inaccessible areas is limited
- Aerovision provides a low-cost, modular UAV platform with real-time vision and sensor data
- Enables fast decision-making for disaster management, surveillance, and environmental monitoring.

INTRODUCTION

- Aerovision is a multi-purpose UAV system integrating flight control, sensors, and AI-based vision
- Designed to perform aerial surveillance, mapping, and human/environment detection
- Focuses on affordability, modularity, and ease of deployment for societal application

KEY FEATURES

- Autonomous UAV-based aerial monitoring
- Real-time image and sensor data acquisition
- AI-based human/object detection
- Modular sensor integration (environmental + vision)
- Low-cost and scalable system architecture
- Suitable for disaster response and large-area surveys

ARCHITECTURE

- Coustom integrated circuit.
- Camera module for live aerial vision
- Environmental sensors for data collection
- A processing for detection and analysis
- Wireless data transmission to ground station
- Modular design allows easy upgrades and customization

RESULTS

- Successful UAV flight with stable control
- Real-time aerial data and video acquisition achieved
- Demonstrated feasibility of low-cost intelligent aerial monitoring
- Provided reliable data for further AI-based analysis

FUTURE SCOPE

1. **Integration of advanced AI models for higher accuracy**
2. **Swarm-based multi-UAV deployment**
3. **Long-range communication and cloud connectivity**

HYBRID SOLAR POWERED RC PLANE

PROJECT NAME:	Hybrid Solar Powered RC Plane
PROJECT GROUP:	Kunal Pisudde

WHY THIS PROJECT ?

- Conventional RC aircraft have limited flight endurance
- Inefficient power utilization causes rapid battery discharge
- Direct battery supply leads to voltage mismatch and energy losses
- Need for energy-efficient and sustainable UAV systems
- Increasing demand for long-duration aerial missions such as monitoring and surveillance

INTRODUCTION

- This project focuses on extending the flight time of an RC aircraft
- Implements an efficient energy management system
- Integrates solar cells, Li-Po battery, voltage regulation and power control
- Ensures stable and reliable power delivery under varying flight conditions
- Improves overall efficiency, endurance and battery life

KEY FEATURES

- Hybrid solar-assisted power system
- High-efficiency buck–boost DC–DC voltage regulator
- Battery protection against:
  1. Overcharging
  2. Over-discharging
  3. Excess current
- Lightweight and compact system
- Improved flight endurance and reliability

ARCHITECTURE

- Solar panels convert sunlight into electrical energy
- Voltage Regulator Module:
  - Steps up/down voltage automatically
  - Maintains constant output
- Li-Po battery stores energy and provides backup power
- ESC regulates motor speed efficiently
- BLDC motor converts electrical energy into thrust
- Ensures safe and efficient energy utilization

RESULTS

- Significant increase in flight duration
- Stable voltage supply during changing load conditions
- Reduced battery stress and discharge rate
- Improved system efficiency compared to conventional RC aircraft
- Reliable and smooth flight performance achieved

FUTURE SCOPE

1. **Use of advanced and lightweight solar panels**
2. **Implementation of smart energy management algorithms**
3. **Development of fully sustainable aerial platforms**

TOROIDAL PROPELLERS

PROJECT NAME:	Design And Development Of Toroidal Propellers
PROJECT GROUP:	Deep Goje, Tejas Kale, Kunal Pisudde, Manthan Kolte

WHY THIS PROJECT ?

- Conventional propellers generate high aerodynamic losses
- High aerodynamic losses reduce propulsion efficiency
- Excessive noise affects urban drone applications
- Vibrations reduce structural life of UAV systems
- Need for efficient and low-noise propulsion technology

INTRODUCTION

- Toroidal propeller uses closed-loop blade geometry
- Reduces airflow disturbances and drag
- Improves airflow circulation and thrust efficiency
- Enhances stability and aerodynamic performance
- Suitable for drones and advanced UAV platforms

KEY FEATURES

- Closed-loop toroidal blade structure
- Reduced tip losses
- Improved thrust-to-power ratio
- Lower acoustic noise emission
- Reduced mechanical vibrations
- Enhanced flight stability

ARCHITECTURE

- Ring-shaped blades improve airflow and balance thrust (refer figure)
- Continuous surface reduces drag and boosts efficiency
- Uniform pressure ensures smooth and stable flight
- Toroidal loop lowers energy loss and vibrations
- Optimized airflow channels reduce noise
- Closed-loop design enhances stability and safety

RESULTS

- Increased propulsion efficiency allows drones to fly longer with less power
- Reduced noise emission makes the drone quieter in urban environments
- Improved thrust performance enhances lift and maneuverability
- Lower mechanical vibrations protect drone structure and electronics
- Enhanced flight stability ensures smoother and safer operation

FUTURE SCOPE

1. **Can be adapted for larger drones and UAVs**
2. **Potential for energy-efficient delivery drones**
3.

AERIAL INSIGHTS

PROJECT NAME:	Aerial Insight- Dam Survey
PROJECT GROUP:	Tejas Kale

WHY THIS PROJECT ?

- To understand the use of UAV technology for infrastructure inspection.
- To detect cracks and structural damage in dams safely and efficiently.
- To reduce manual inspection risks in hard-to-reach areas.
- To gain practical experience in drone surveying and data collection.
- To explore real-world engineering applications of drones.

INTRODUCTION

- Used UAV (drone) technology for inspection and monitoring of dam structures.
- Aimed to detect cracks and structural defects safely and efficiently.
- Aerial survey allows inspection of hard-to-reach areas without manual risk.
- Helps in early detection of damage and preventive maintenance.
- Provides accurate visual data for structural analysis.

KEY FEATURES

- UAV-Based Inspection – Drone used for safe and efficient dam monitoring.
- High-Resolution Imaging – Captures clear images for crack detection.
- Access to Difficult Areas – Inspects locations that are hard to reach manually.
- Fast Survey Process – Covers large dam areas in less time.
- Accurate Visual Data – Provides reliable information for structural analysis.

ARCHITECTURE

- UAV Platform – Drone used to fly over the dam and collect data.
- Camera System – Captures images and videos of dam surfaces.
- Control System – Transmitter and flight controller used to operate the UAV.
- Data Storage – Images and videos stored for inspection and study.
- Analysis Method – Collected data used to identify cracks and defects.

RESULTS

- Successfully conducted aerial survey of the dam using UAV technology.
- Captured clear images and videos for crack detection.
- Identified visible cracks and surface defects in the structure.
- Achieved safe and efficient inspection without manual risk.
- Gained practical experience in drone surveying and inspection techniques.

FUTURE SCOPE

1. **Integrate thermal or advanced cameras for deeper structural analysis.**
2. **Apply UAV inspection to bridges, buildings, and other strutures.**
3. **Develop automated flight paths for regular dam monitoring.**

MINI DRONE

PROJECT NAME:	Mini Drone
PROJECT GROUP:	Tejas Kale

WHY THIS PROJECT ?

- To provide hands-on experience in flying and controlling drones.
- Helps students practice basic aerial maneuvers safely.
- Teaches how gyro stabilization and flight controls work.
- Useful for learning indoor and outdoor piloting skills.

INTRODUCTION

- This project involves using a mini quadcopter for learning and piloting purposes.
- The drone has a 6-axis gyro for stable flight and easy handling.
- Features like one-key return, LED lights, and multiple flight modes make it suitable for practice and skill development.
- The drone provides a safe platform for beginners to understand flight control and basic navigation.

KEY FEATURES

- 6-axis gyro for stable flight
- One-key return for easy navigation
- LED lights for night flying
- Multiple flight speeds: slow, medium, high
- 360° rolling and accurate localization
- Suitable for indoor and outdoor practice

ARCHITECTURE

- The remote control sends signals to the drone via 2.4GHz frequency.
- The 6-axis gyro stabilizes the drone during flight.
- Motors and propellers respond to commands for ascent, descent, forward/backward, and side movements.
- Features like one-key return, LED lights, and 360° rolling actions enhance usability and learning experience.

RESULTS

- Users successfully learned to pilot the drone both indoors and outdoors.
- Improved understanding of flight control, stability, and navigation.
- Practiced maneuvers like ascent, descent, forward/backward movement, and 360° rolling.

FUTURE SCOPE

1. **Can be used to practice advanced flight techniques and obstacle navigation.**
2. **Useful for learning drone photography and aerial mapping basics.**
3. **Can help train beginners for handling larger or professional drones.**

RC CAR

PROJECT NAME:	Design And Development Of RC Car
PROJECT GROUP:	Deep Goje, Yuraj Harne, Tejas Kale

WHY THIS PROJECT ?

- To learn and experiment with designing a custom RC chassis using aluminium pipes.
- To gain hands-on experience in mechanical assembly and material handling.
- To prepare for RC competitions by testing stability, performance, and durability.
- To encourage creativity and innovation by building something ourselves instead of using pre-made parts.

INTRODUCTION

- Designed and built a custom RC chassis using aluminium pipes.
- Explored a new type of chassis made entirely by ourselves.
- Learned about mechanical design, material selection, and structural stability.
- Prepared for RC competitions by testing performance and durability.

KEY FEATURES

- Custom Aluminium Chassis: Built from scratch for strength and lightweight design.
- Durable Structure: Aluminium pipes provide stability during RC operation.
- Modular Design: Easy to assemble, modify, or repair for experiments or competitions.
- Competition-Ready: Designed to perform in RC racing or obstacle challenges.
- Hands-On Learning: Focused on practical mechanical skills and material handling.

ARCHITECTURE

- Metallic Frame: Custom aluminium chassis forming the main structure.
- BLDC Motors: Provide efficient and high-speed propulsion for the RC vehicle.
- Transmitter & Receiver: Enable remote control communication for movement and navigation.
- Power Distribution Board: Distributes electrical power efficiently to all components.
- ESC & LiPo Battery: Electronic Speed Controllers regulate motor speed; LiPo battery supplies high-energy power.

RESULTS

- Successfully designed and built a custom RC chassis using aluminium pipes.
- Vehicle was stable, durable, and functional for RC operation.
- Gained hands-on experience in mechanical design, assembly, and electronics integration.
- Tested in RC competition scenarios, validating performance and modular design.

FUTURE SCOPE

1. ☐ **Explore lightweight composites or alloys for better performance.**
2. ☐ **Add sensors for autonomous navigation or obstacle detection.**
3. ☐ **Upgrade motors, ESC, or battery for higher speed and efficiency.**

RC CAR

PROJECT NAME:	RC Car By Composite Sheet
PROJECT GROUP:	Deep Goje, Yuraj Harne, Tejas Kale

WHY THIS PROJECT ?

- To learn chassis design using composite materials.
- To make a lightweight and strong RC structure.
- To gain hands-on experience in fabrication and assembly.
- To test performance for RC competitions.
- To improve practical engineering skills.

INTRODUCTION

- Designed and built an RC car chassis using composite sheet material.
- Aim was to explore lightweight and strong materials for RC vehicle design.
- Components like motor, battery, and electronics were mounted on the composite chassis.
- Helped understand material properties and structural strength.
- Project was useful for learning and RC competition preparation.

KEY FEATURES

- Composite Sheet Chassis – Lightweight and strong structure.
- Custom Design – Chassis designed and cut manually.
- Component Mounting – Motor, battery, and electronics mounted securely.
- Lightweight Structure – Improves speed and efficiency.
- Competition Suitable – Designed for RC testing and competitions.

ARCHITECTURE

- Composite Chassis – Base structure supporting all components.
- Motor System – Provides movement to the RC car.
- Drive System – Transfers power to the wheels.
- Control System – Receiver and transmitter control movement.
- Power System – Battery supplies electrical energy.

RESULTS

- Successfully built an RC car chassis using composite sheet.
- Achieved a lightweight and stable structure.
- Components were mounted and tested successfully.
- Improved understanding of chassis design and materials.

FUTURE SCOPE

1. **Use stronger composite materials like carbon fiber.**
2. **mprove chassis design for better strength and stability**
3. **Add sensors for autonomous navigation or obstacle detection.**

RC CAR (CHASSIS)

PROJECT NAME:	Design And Development Of RC Car
PROJECT GROUP:	Kunal Pisudde, Manthan Kolte , Rushikesh Payghan, Gaurav Zade

WHY THIS PROJECT ?

- To gain practical knowledge of RC vehicle mechanics, electronics, and control systems.
- To prepare for RC competitions, testing speed, stability, and handling.
- To understand component interaction, troubleshooting, and real-te system performance.
- To enhance engineering skills like problem-solving, teamwork, and system integration.

INTRODUCTION

- Used a ready-made RC chassis frame as the base structure.
- Mounted and integrated all components like motors, electronics, battery, and drivetrain ourselves.
- Learned how mechanical and electronic systems work together in an RC car.
- Gained hands-on experience in assembly, testing, and troubleshooting.
- Prepared for competitions by ensuring performance, stability, and control.

KEY FEATURES

- Ready-made nylon chassis frame with roll cage and front bumper for durability.
- Four-link suspension with oil-filled shock absorbers for smooth handling.
- Two-speed transmission and high-torque motor for versatile performance.
- Metal-geared differentials and universal drive shafts for efficient power transfer.
- ESC, servos, LiPo battery, and 2.4GHz transmitter for precise electronic control.

ARCHITECTURE

- Metallic/Nylon Chassis Frame: Base structure that supports and holds all components together.
- Motor & Transmission System: High-torque motor and two-speed transmission provide motion and power to the wheels.
- Drive System: Differentials, drive shafts, and portal axles transfer power efficiently to the wheels.
- Control Electronics: ESC, receiver, and servos control speed, steering, and gear shifting.
- Power System: LiPo battery and power distribution provide electrical energy to all components.

RESULTS

- Successfully mounted and integrated all components on the ready-made chassis.
- RC car performed smoothly with good control and stability.
- Gained practical experience in assembly, testing, and troubleshooting electronics and mechanical parts.
- Prepared for RC competitions, understanding how design and component choices affect real-world performance.

FUTURE SCOPE

1. **Refine design to improve handling and stability in competitions.**
2. **se lightweight composites or alloys for improved performance.**
3. - **Upgrade motors, ESC, or battery for higher speed and efficiency.**

DESIGN AND DEVELOPMENT OF ESP32

PROJECT NAME:	Design And Development OF ESP32
PROJECT GROUP:	Sahil Khandare

WHY THIS PROJECT ?

- o learn the process of designing a custom PCB for ESP32.
- To understand the connections and working of the ESP32 microcontroller.
- To gain practical knowledge of circuit design and PCB layout.
- To develop a compact and reliable ESP32 development board.
- To use the board for future embedded and IoT projects.

INTRODUCTION

- This project focuses on the design and development of a custom ESP32-based PCB.
- The PCB was designed to provide proper connections for power supply, programming, and input/output pins of the ESP32.
- Components were mounted on the board to make a compact and functional system.
- The project helped in understanding PCB design, routing, and hardware development.
- The developed board can be used for embedded systems and IoT applications.

KEY FEATURES

- Custom designed PCB for ESP32 module.
- Compact and organized component layout.
- Proper power supply connections for stable operation.
- Accessible input and output pins for easy interfacing.
- Suitable for embedded and IoT projects.

ARCHITECTURE

- ESP32 module acts as the main controller.
- Power supply section provides required voltage to the ESP32.
- Input and output pins are connected through PCB tracks.
- Programming interface is provided for uploading code.
- All components are mounted on a single PCB.

RESULTS

- he ESP32 custom PCB was successfully designed and developed.
- All components were properly mounted and connected.
- The board powered on and functioned correctly.
- The project improved understanding of PCB design and ESP32 hardware.

FUTURE SCOPE

1. **The board can be improved with better power regulation.**
2. **The design can be made more compact and efficient.**
3. **The board can be improved with better power regulation.**

GIMBAL

PROJECT NAME:	Gimblal Drone Testing Stand
PROJECT GROUP:	Kunal Pisudde, Atharva Raut, Pranav Harne

WHY THIS PROJECT ?

- To provide a stable platform for testing drone gimbals.
- Mild steel gives strong and durable support.
- Helps in safe testing without flying the drone.
- Useful for checking gimbal movement and performance.

INTRODUCTION

- A gimbal drone testing stand is used to hold the drone securely during testing and adjustment of the gimbal system.
- It provides a stable platform for checking gimbal movement and drone performance safely without actual flight.

KEY FEATURES

- Stable support for drone testing
- Safe testing without flight
- Easy mounting of drone
- Smooth gimbal movement testing

ARCHITECTURE

- The testing stand provides a stable and secure platform where the drone can be mounted safely.
- Once mounted, the gimbal system of the drone is tested and observed to ensure proper movement and performance.
- Any necessary adjustments can be made while the drone remains on the stand, without the need for actual flight.

RESULTS

- Drone was securely mounted for testing.
- Gimbal movement and performance were successfully observed.
- Adjustments were made safely without flight.

FUTURE SCOPE

1. **Can be used to test gimbal movements like roll, pitch, yaw, and throttle.**
2. **Can help in calibrating and improving gimbal performance.**
3. **Allows repeated testing of drones in a controlled environment.**

RC PLANE

PROJECT NAME:	RC Plane (PLA Based 3D Printed)
PROJECT GROUP:	Tejas Kale, Kunal Pisudde, Rushikesh Payghan, Manthan Kolte

WHY THIS PROJECT ?

- 3D printing allows lightweight and customizable aircraft design.
- PLA material makes the structure easy to manufacture.
- Helps in understanding aerodynamics and aircraft structure.
- Useful for learning RC aircraft design and development.

INTRODUCTION

- The PLA-based 3D printed RC plane is designed and developed using lightweight printed parts and electronic components for controlled flight.
- The project demonstrates aircraft design, electronics integration, and remote-controlled operation using a simple and cost-effective structure.

KEY FEATURES

- Lightweight PLA 3D printed structure
- Aerodynamic airframe design
- Remote controlled operation
- Easy assembly and repair

ARCHITECTURE

- 3D printed PLA parts form the aircraft structure.
- Electronic components like motor, ESC, battery, and receiver are installed.
- The transmitter sends control signals to the receiver.
- The motor and servos control the movement of the plane.
- The plane flies based on remote control inputs.

RESULTS

- Enhanced flight performance and stability
- Improved piloting skills and control accuracy
- Safe and smooth flight operations
- Increased exploration and experimentation capability
- Better understanding of aerodynamic principles in practice

FUTURE SCOPE

1. **Use of lightweight composite materials to improve strength and durability**
2. **Add sensors to monitor flight**
3. **Carry small payloads like cameras**

3D PRINTED RC PLANE

PROJECT NAME:	3D Printed RC Plane
PROJECT GROUP:	Sahil Khandare

WHY THIS PROJECT ?

- To showcase a new RC plane design for exhibition purposes.
- To demonstrate the use of 3D printing in aircraft development.
- To test new materials for lightweight aircraft structures.
- To explore innovative RC plane design and construction methods.

INTRODUCTION

- This project presents a 3D printed RC plane developed using a new design and material for exhibition and demonstration purposes.
- The airframe was designed using CAD software and manufactured using 3D printing to create a lightweight and innovative aircraft structure.

KEY FEATURES

- New 3D printed aircraft design
- Lightweight printed structure
- Use of new material
- Modular parts for easy assembly

ARCHITECTURE

- RC plane parts designed using CAD software.
- Parts manufactured using 3D printing with new material.
- Electronic components installed inside the airframe.
- Transmitter and receiver used for flight control.
- Motor and servos control the movement of the plane.

RESULTS

- 3D printed RC plane was successfully developed.
- New design and material worked effectively.
- Plane was successfully displayed for exhibition.

FUTURE SCOPE

1. **Testing of improved designs and materials.**
2. **Development of lighter and stronger airframes.**
3. **Use for research and advanced RC plane projects.**

RECYCLING OF LI-PO BATTERY

PROJECT NAME:	Recycling of Li-Po Battery
PROJECT GROUP:	Rushikesh Payghan, Manthan Kolte

WHY THIS PROJECT ?

- Li-Po batteries contain valuable materials that can be reused.
- Improper disposal may cause fire and safety hazards.
- Recycling helps reduce electronic waste.
- Faulty 4S Li-Po batteries can be repaired by replacing only the damaged cell instead of discarding the whole battery pack.

INTRODUCTION

- LiPo batteries are widely used in drones and RC models.
- A 4S battery pack may fail due to one damaged cell.
- This project focuses on testing batteries and reusing good cells.
- Faulty cells were replaced to make a working battery pack.

KEY FEATURES

- Li-Po Battery Testing
- Faulty cell identification
- Cell replacement method
- Battery reuse
- Safe handling

ARCHITECTURE

- Two 4S Li-Po batteries selected
- Good and faulty cells compared
- Good cell transferred to Battery
- Final working battery pack obtained

RESULTS

- Faulty cells were successfully identified.
- One working 4S battery was made from two batteries.
- Battery reuse reduced waste and improved safety.

FUTURE SCOPE

1. **Testing of more Li-Po batteries**
2. **Improved battery repair methods**
3. **Large-scale battery recycling**

OBSTACLE AVOIDANCE CAR

PROJECT NAME:	Obstacle Avoidance Car
PROJECT GROUP:	Gaurav Ingle

WHY THIS PROJECT ?

- Helps understand basic robotics and automation.
- Demonstrates automatic obstacle detection.
- Reduces chances of collision.
- Useful for autonomous robot applications.

INTRODUCTION

- The obstacle avoidance car is an autonomous robotic vehicle that uses an ultrasonic sensor to detect obstacles in its path and automatically changes direction with the help of an Arduino-controlled motor system.
- This project demonstrates the use of sensors, microcontrollers, and motor drivers to build a smart vehicle that can move safely without human control by avoiding collisions with nearby object.

KEY FEATURES

- Uses an ultrasonic sensor to measure distance and detect obstacles in front of the car.
- Arduino Uno controls the system and processes sensor data for decision making.
- Motor driver module controls the speed and direction of the DC motors.
- The car automatically changes direction when an obstacle is detected.

ARCHITECTURE

- The ultrasonic sensor continuously measures the distance between the car and nearby obstacles.
- The sensor sends distance data to the Arduino Uno for processing.
- The Arduino analyzes the data and decides the direction of movement.
- The motor driver module receives signals from the Arduino and controls the DC motors.
- The car moves forward or changes direction to avoid obstacles automatically.

RESULTS

- Obstacles were detected successfully.
- Car avoided collisions automatically.
- Smooth autonomous movement achieved.

FUTURE SCOPE

1. **Use advanced sensors for better detection**
2. • **Wireless control system**
3. **Application in smart robots and vehicles**